MASTER'S THESIS
(PHYSICS)

Evolution of Open Quantum Systems under Continuous Measurement of their Environment

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

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CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys).
My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.
It aims to study from an academical perspective at first, the subject of continuous measurement considering open quantum systems. The main objective was to explore a subject almost-unknown in details to either my supervisor or myself. We wanted to understand the possibilities and how to handle this domain properly. This work could then lead to other searches linked to continuous measurements.

By familiarizing ourselves with the topic, we decided to adopt this perticular frame of report. The structure of this document is meant to help any unfamiliar reader (about continuous measurement), to discover and explore its fundamentals. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts.

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement, instead, position is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. The definition of trajectory will therefore be addressed, but one question remains paramount when mentioning systems: how can we even define a quantum system properly?

In the ideal case, systems are treated as closed, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2] However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. In doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Other useful theories have been developed through the last decades, allowing quantum mechanics' description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where the concept of measurement is fully described. Measuring a system has a tremendous impact on the its quantum properties. Furthermore, since any experimental probe inevitably interacts with the system, it is crucial to figure out the nature of this interaction. Measuring a system without impacting its quantum properties leads to the concept of weak measurements, that we will develop further below.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties. For example it can lose its entanglement[1]. Along the same lines, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment, this corresponds formally to a superposition of 'alive' and 'dead' states prior to measurement. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" in the state measured and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

Practically, this work will be organized as follows: the first section covers an academical progression from basics in quantum mechanics up to the aforementioned theories. A second section will cover the photodetection and homodyne detection of open quantum systems. The qubit and harmonic oscillator will be used to illustrate our point. The final part is the study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow[\text{in space A}]{\text{measuring 0}} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

I From closed systems to continuous measurements — in a nutshell

In this section we will briefly summarize the definitions needed to our study.

I.i) Generalities on quantum mechanics

I.i.1⟩ States and time evolution

In quantum mechanics a physical system's state is described by a vector in a Hilbert space:

$$|\psi\rangle \in \mathcal{H} \tag{1}$$

The evolution of a system depends on its energy, the Hamiltonian H is an hermitian[2] operator that accounts for the total energy of a system. The Schrödinger equation states it explicitly:

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle \tag{2}$$

Instead of looking at the state at time t we can use an unitary[3] operator $\hat{U}$ such that, $|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle$. From now on, we will omit the $\hat{}$ symbol over operators when it is obvious. We get,

$$\dot{U}(t,t_0) = -\frac{i}{\hbar} H(t) U(t,t_0) \tag{3}$$

One can solve this differential equation in general by using a chronological time-ordering operator [3]. Giving for the state evolution,

$$|\psi(t)\rangle = \mathcal{T}_{\leftarrow} \exp\left[-\frac{i}{\hbar}\int_{t_0}^{t} H(s)\mathrm{d}s\right]|\psi(t_0)\rangle \tag{4}$$

Or, when the Hamiltonian is time-independent,

$$|\psi(t)\rangle = \underbrace{e^{-\frac{iH(t-t_0)}{\hbar}}}_{U(t,t_0)}|\psi(t_0)\rangle \tag{5}$$

This evolution of the state with respect to time is said **unitary and reversible** because U is unitary, and, because this equation also describe the time reversed evolution $t \to t_0$: $|\psi(t_0)\rangle = U^\dagger(t,t_0)\,|\psi(t)\rangle$.

I.i.2⟩ Observables

Any physical quantity is represented in quantum mechanics by an **observable**: an operator whose eigenvectors form an orthonormal basis of the Hilbert space $\mathcal{H}$ [2] and whose eigenvalues are positive to represent a physical measurement. For finite dimension Hilbert space, hermitian operators[4] are observables. In the following we will consider finite dimension, non-degenerate cases to illustrate but one can generalize [2].

[2] $\hat{H}^\dagger = \hat{H}$, where † is the conjugate transpose operation, defining the hermitian adjoint.

[3] $UU^\dagger = U^\dagger U = \mathbb{I}$, where $\mathbb{I}$ is the identity.

[4] Hermitian operators always form a basis in finite dimension, we can use Gram-Schmidt process to find its orthonormal counterpart and being self-adjoint implies having real eigenvalues

A natural operator whose eigenbasis is particularly interesting is the Hamiltonian, its eigenvalues correspond to the possible energy levels of a system.
Let us denote $I_{\mathcal{H}} = [\![1; \dim(\mathcal{H})]\!]$ throughout this document.

$$\forall n \in I_{\mathcal{H}}, \quad H\left|\phi_n\right\rangle = E_n\left|\phi_n\right\rangle \tag{6}$$

Let us call $\{\phi_n\}_{n\in I_{\mathcal{H}}}$ the family of orthonormal eigenvectors of H which forms a basis of $\mathcal{H}$.
Since it forms a basis of the Hilbert space, one can describe any state in this very basis:

$$\forall \left|\psi\right\rangle \in \mathcal{H}, \qquad \left|\psi\right\rangle = \sum_{n\in I_{\mathcal{H}}} \underbrace{c_n}_{\langle\phi_n|\psi\rangle} \left|\phi_n\right\rangle \tag{7}$$

This kind of description in a base is obviously valid for any observable, not only for the Hamiltonian. We can complete this part with the normalization of the state, for proper handling of scalar products, we need to set $|\psi| = \sqrt{\langle\psi|\psi\rangle} = 1$. The condition $\langle\psi|\psi\rangle = \sum_n |c_n|^2 = 1$ is thus equivalent.

I.i.3⟩ Purity of states

Taking the qubit example, let $\alpha, \beta \in \mathbb{C}$ to write the qubit state as a superposition of excited/ground state as:

$$\left|\psi\right\rangle = \alpha\left|e\right\rangle + \beta\left|g\right\rangle \qquad \text{where, } |\alpha|^2 + |\beta|^2 = 1 \tag{8}$$

By getting ahead of QMT, we can introduce the probability to find a system in one of the possible eigenstates of an observable. Let us take the following example, what is the probability to find the energy E_n, defined by Eq. (6), from a system in $\left|e\right\rangle$ state? From a system in $\left|g\right\rangle$ state?

$$\mathbb{P}_e[E_n] = |\langle\phi_n|e\rangle|^2 \tag{9}$$
$$\mathbb{P}_g[E_n] = |\langle\phi_n|g\rangle|^2 \tag{10}$$

A physical state can be, by the concepts of quantum mechanics, a mixture of statistical probabilities [2]. Let us now consider back the superposition from Eq. (8), a physical mixture would be described by mixing states $\left|e\right\rangle, \left|g\right\rangle$ with weights $|\alpha|^2, |\beta|^2$:

$$\mathbb{P}_\psi[E_n] \stackrel{?}{=} |\alpha|^2\ \mathbb{P}_e[E_n] + |\beta|^2\ \mathbb{P}_g[E_n] \tag{11}$$

In fact, quantum mechanics gives the exact value of this probability:

$$\mathbb{P}_\psi[E_n] = |\langle\phi_n|\psi\rangle|^2 \tag{12}$$
$$= |\alpha|^2|\langle\phi_n|e\rangle|^2 + |\beta|^2|\langle\phi_n|g\rangle|^2 + 2\,\mathrm{Re}\left\{\alpha\beta^*\,\langle\phi_n|e\rangle\,\langle\phi_n|g\rangle^*\right\} \tag{13}$$

The mixture term appears but the impact of cross terms also appear in the real probability to find the superposition in one of the eigenstates. This notion of hidden correlations due to superposition has been highlighted by a new formalism, **density matrices**.
Basically, instead of a ket $\left|\psi\right\rangle$ we describe a system by a density matrix ρ which, for **pure** systems is simply:

$$\rho_{\mathrm{p}} = \left|\psi\right\rangle\left\langle\psi\right| \tag{14}$$

The density matrix is thus an operator of Hilbert space elements[5] $\rho : \mathcal{H} \to \mathcal{H}$. For a **statistical mixture**, thus non-pure, the density matrix in a basis $\{\phi_n\}_{n \in I_{\mathcal{H}}}$ takes the form:

$$\rho_{\text{sm}} = \sum_{n \in I_{\mathcal{H}}} \mathbb{P}_n \left|\phi_n\right\rangle\!\left\langle\phi_n\right| \tag{15}$$

The components of the matrix are precisely, on the diagonal, the probabilities to find the state of the system in one of the eigenstates of the observable. The off-diagonal components are called **coherences**, they represent interference effects between the eigenstates, they can appear when the system's state is a linear coherent superposition of these eigenstates [2].
The mathematical definition of a density matrix is in any case [2, 3]:

- ρ is hermitian $\rho^\dagger = \rho$
- ρ is semi-definite $\forall \left|\alpha\right\rangle \in \mathcal{H}, \quad \left\langle\alpha\right| \rho \left|\alpha\right\rangle \geq 0$
- ρ respects $\text{Tr}\left[\rho\right] = 1$

I.i.4⟩ Postulates of quantum mechanics

Physicists developed a corpus of the main properties of quantum mechanics, here are summarized the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathcal{L}(\mathcal{H})$. For pure states we can describe them using the Dirac formalism $\left|\psi\right\rangle \in \mathcal{H}$, or using the pure matrix density formalism $\rho_p = \left|\psi\right\rangle\!\left\langle\psi\right|$.

2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.

3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues a of $\hat{A}$, defined by $\hat{A}\left|a\right\rangle = a\left|a\right\rangle$.

4. When measuring $\mathcal{A}$ on a normalized state, we know the probability to get one of the possible eigenvalues: $\mathbb{P}(a_n)$ (which depends on the spectrum of the observable).

5. If the measurement of the physical quantity $\mathcal{A}$ on the system gives the result a_n, the state of the system immediately after the measurement becomes the eigenstate $\left|\phi_n\right\rangle$.

6. The time evolution of the wavefunction $\left|\psi(t)\right\rangle$ is governed by the *Schrödinger* equation:
$$\frac{\mathrm{d}\left|\psi(t)\right\rangle}{\mathrm{d}t} = -\frac{i}{\hbar} H(t)\left|\psi(t)\right\rangle$$
For density matrices, the evolution is driven by the *Liouville-von Neumann* equation[6]:
$$\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}\left[H(t), \rho(t)\right] \equiv \mathcal{L}(t)\rho(t) \tag{16}$$

[5] ρ is also an operator of the dual, $\mathcal{H}^*$ because we find that $\left\langle\phi_j\right| \rho \propto \left\langle\phi_j\right|$.
We could even ague that $\rho \in \mathcal{L}(\mathcal{H})\ldots$

[6] $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is the commutator of $\hat{A}$ and $\hat{B}$

I.ii) Open Quantum Systems (OQS)

In the previous part, we saw that one can fully describe a system, observables, states and their evolution. We recall that this description is ideal in the sense that the system is considered to be completely isolated from any interaction with its environment. However, we will introduce in this section the concepts of OQS which lead to a more physical description of quantum mechanics.

It is important to understand that the total system is a closed system, the total system space is mathematically $\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E$. The total system T thus follows a unitary time evolution, but, it is not the case for the physical system of interest S. A new kind of dynamics will arise for S due to its coupling with E, this is why we call S an open quantum system.

By following the work of *Breuer&Petruccione* [3], we need to jump in a different representation of the equations: the interaction picture. In the *Schrödinger* picture, states evolve with respect to time, observables can be constants or depend explicitly on time. Instead, in the *Heisenberg* picture it is the opposite, observables carry the whole dynamics and states are constants (see Eq. A. 2). The interaction picture is a mix of both, where states and observables are functions of time.

Interaction picture begins with the perturbation theory, where we split the Hamiltonian in two parts, the free Hamiltonian H_0 and the interaction Hamiltonian H_I

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I \tag{17}$$

$$U(t, t_0) = U_0(t, t_0) U_I(t, t_0) \tag{18}$$

We define the interaction Hamiltonian by,

$$E_n(t) = U_0^\dagger(t) E_n U_0(t) \tag{19}$$

$$H_I(t) = \sum_{n \in I_{\mathcal{H}}} \overbrace{|n\rangle\langle n|}^{A_n} \otimes E_n(t) \tag{20}$$

We can then find the useful following relations:

$$\tilde{\rho}(t) = U_I(t) \rho_S U_I^\dagger(t) = U_0^\dagger(t) \rho_S(t) U_0(t) \tag{21}$$

$$\tilde{O}(t) = U_0^\dagger(t) O_S(t) U_0(t) \tag{22}$$

One can then find the equation that governs the dynamics of ρ_S[7] in the *Schrödinger picture* by making progress in the *interaction picture*. We call this equation the *Redfield Master Equation* but its expression is analytically formal. To go further, one needs to make important assumptions known as the *Born-Markov* approximations and the *secular* approximation.

[7] Here and in the following, ρ_S is the state of the system obtained by tracing out the environnement from the total state: $\rho_S = \mathrm{Tr}_E[\rho]$

♦ The *Born* approximation consists in saying that the environment acts as a thermodynamical reservoir, *i.e.*, the environment is so big compared to the system that the interaction does not alter its state:

$$\tilde{\rho}_T(t) = \tilde{\rho}_S(t) \otimes \rho_E \tag{23}$$

It describes a so-called "weak-coupling" between the system and the environment.

♦ The *Markov* approximation assures that the entire system has no memory. That is to say, the environment is so rapidly agitating that it dissipates information of the system instantaneously. This implies that the environment comes back to equilibrium instantaneously after each interaction. Physically, the timescale of the reservoir correlation function decay τ_E is smaller than the timescale τ_R on which the system varies "appreciably".

These two last approximations lead to the *Markovian quantum master equation.*

♦ The *secular* approximation allows the possibility to neglect rapidly oscillating terms in the quantum *Markov* master equation.

I.iii) Quantum Measurement Theory (QMT)

II Continuously measured quantum systems – obtaining a Stochastic Master Equation (SME)

A Representations of quantum mechanics

Complement of quantum mechanics An average is defined in quantum mechanics by:

$$\begin{aligned}\langle O\rangle_\psi &= \langle\psi|O|\psi\rangle\\ \langle O\rangle_\rho &= \mathrm{Tr}\,[\rho O]\end{aligned}$$

We can see the two definitions are strictly equivalent, to illustrate let us pick a pure state $\rho = |\psi\rangle\langle\psi|$ and the base of an Hamiltonian $\{\phi_n\}_{n\in I_\mathcal{H}}$:

$$\begin{aligned}\langle O\rangle_\rho &= \mathrm{Tr}\,[\rho O]\\ &= \mathrm{Tr}\,[|\psi\rangle\langle\psi|\,O]\\ &= \mathrm{Tr}\,[\,\langle\psi|O|\psi\rangle]\end{aligned}$$

However,

$$\langle\psi|O|\psi\rangle \in \mathbb{C}$$

Thus,

$$\langle O\rangle_\rho = \langle\psi|O|\psi\rangle$$

□

A.a) Schrödinger and Heisenberg pictures

In this appendix we will work by denoting the pictures with their acronyms, but we chose to lighten the notation of the total state $\rho_T \equiv \rho$ here. Please pay attention not to misinterpret S in this part means *in the Schrödinger picture*, whereas, S in the main part means *state of the system S*.

In the *Schrödinger* picture we saw that $|\psi(t)\rangle = U(t)\,|\psi\rangle$ where, $U(t) = e^{-\frac{i}{\hbar}Ht}$ in the case of a time-independent Hamiltonian with $t_0 = 0$. The following demonstration can be made in the general case where the Hamiltonian is time dependent using Eq. (4).

If we consider the average value of an observable $A_S(t)$ in the *Schrödinger* picture,

$$\begin{aligned}\langle A_S(t)\rangle &= \langle\psi(t)|A_S|\psi(t)\rangle\\ &= \langle\psi|\underbrace{U^\dagger(t)A_S(t)U(t)}_{A_H(t)}|\psi\rangle\end{aligned} \tag{A. 1}$$

This naturally let appear the *Heisenberg representation* of observables and states.
Indeed, we have $\forall t,\ |\psi(t)\rangle_H = |\psi(t_0)\rangle_S \equiv |\psi\rangle_H$ which links both pictures at the initial time.
We can link both representation by calculating the evolution of observable in the *Heisen-*

berg picture:

$$\begin{aligned}
\frac{\mathrm{d}A_H}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[U^\dagger A_S U\right] \\
&= \frac{\mathrm{d}U^\dagger}{\mathrm{d}t}A_S U + U^\dagger \frac{\partial A_S}{\partial t} U + \frac{\mathrm{d}U}{\mathrm{d}t} A_S U \\
&= \frac{i}{\hbar} U^\dagger H_S A_S U + U^\dagger \frac{\partial A}{\partial t} U - \frac{i}{\hbar} U^\dagger A_S H_S U \\
&= \frac{i}{\hbar}[U^\dagger H_S \overbrace{UU^\dagger}^{\mathbb{I}} A_S U - U^\dagger A_S \overbrace{UU^\dagger}^{\mathbb{I}} H_S U] + \frac{\partial A_S}{\partial t} \\
&= \frac{i}{\hbar}\left[H_H A_H - A_H H_H\right] + \frac{\partial A_S}{\partial t}
\end{aligned}$$

So,

$$\boxed{\frac{\mathrm{d}A_H}{\mathrm{d}t} = \frac{i}{\hbar}\left[H_H, A_H\right] + \frac{\partial A_S}{\partial t}} \tag{A. 2}$$

This link can even lead to the famous *Ehrenfest theorem* for quantum mechanics:

$$\boxed{\frac{\mathrm{d}\left\langle A\right\rangle}{\mathrm{d}t} = \frac{i}{\hbar}\left\langle [H(t), A(t)]\right\rangle + \left\langle \frac{\partial A}{\partial t}\right\rangle} \tag{A. 3}$$

We observe that if the observable doesn't depend explicitly on time (thus is constant), then the dynamics of the operator is fully describe by the *Heisenberg representation*.

A.b) Interaction picture

As mentioned in the main part, see I.ii), we use perturbation theory to describe the interaction between the system S and the environment E:

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I \tag{17}$$

$$U(t, t_0) = U_0(t, t_0) U_I(t, t_0) \tag{18}$$

Recall: $\left|\psi(t)\right\rangle_S = U(t)\left|\psi\right\rangle_S$

$$\rho_S(t) = U(t)\rho_S U^\dagger(t) \tag{A. 4}$$

If we now try to calculate the average (in $\mathcal{H}_T$) of an operator in the *Schrödinger picture*, we can find a new interesting relation:

$$\begin{aligned}
\langle O_S(t)\rangle &= \mathrm{Tr}\left[\rho_S(t) O_S(t)\right] \\
&= \mathrm{Tr}\left[U(t)\rho_S U^\dagger(t) O_S(t)\right] \\
&= \mathrm{Tr}\left[U_0(t) U_I(t)\rho_S U_I^\dagger(t) U_0^\dagger(t) O_S(t)\right] \\
&= \mathrm{Tr}\left[\underbrace{U_I(t)\rho_S U_I^\dagger(t)}_{\tilde{\rho}(t)}\ \underbrace{U_0^\dagger(t) O_S(t) U_0(t)}_{\tilde{O}(t)}\right]
\end{aligned}$$

We can thus define in the interaction picture:

$$\tilde{\rho}(t) = U_I(t)\rho_S U_I^\dagger(t) \tag{21}$$

$$\tilde{O}(t) = U_0^\dagger(t) O_S(t) U_0(t) \tag{22}$$

We can now reverse Eq. (A. 4) and then use Eq. (18):

$$\rho_S = U^\dagger(t)\rho_S(t)U(t) = U_I^\dagger(t)U_0^\dagger(t)\rho_S(t)U_0(t)U_I(t)$$

And, by injecting this into Eq. (21), we get

$$\begin{aligned}\tilde{\rho}(t) &= U_I(t)U_I^\dagger(t)U_0^\dagger(t)\rho_S(t)U_0(t)U_I(t)U_I^\dagger(t)\\ &= U_0^\dagger(t)\rho_S(t)U_0(t)\end{aligned} \tag{21}$$

So that we find the results used in the main part of this work

$$\tilde{\rho}(t) = U_I(t)\rho_S U_I^\dagger(t) = U_0^\dagger(t)\rho_S(t)U_0(t)$$

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